

AKASH JAPE

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EDUCATION

Bachelor of Science in Mechanical Engineering

December 2026

The University of Texas at Austin, Materials Science Concentration, GPA: 3.86/4.00

Relevant Courses: Design Methodology, Machine Elements, Materials Processing, Sketching for Product Design, Dynamic Systems & Controls, Mech. Behavior of Materials, Machine Tool Operations, Mechatronics, Heat Transfer, Fluid Mechanics

EXPERIENCE

SharkNinja – Needham, MA

May 2026 – August 2026

Product Design Engineering Intern, Advanced Development

- Supported prototyping for P1 and P2 builds, focusing on ergonomics, user experience, and assembly design
- Developed a user-operated retention mechanism for a rotating and translating shaft; generated four concept variants and identified leading path via user demos with Product Integrity (PI) and Development (PD) teams
- Prototyped a mechanical safety interlock for a 3-part assembly; incorporated poka-yoke methodology to guarantee correct installation; laser-sintered nylon mechanism validated by PI and PD teams for P2 assembly
- Led product ideation for 2028 workstreams; analyzed desirability-feasibility-viability via competitor benchmarking, culinary R&D, and market research; drafted product pitch and received go-ahead on further development

Qorvo – Richardson, TX

May 2025 – August 2025

Process Engineering Intern

- Applied 5S methodology to design and additively manufacture ESD-compliant thermoplastic tooling; validated improvements through user testing and time-motion studies, reducing excess movement and cycle time
- Led DoE-driven investigation of epoxy dispensing defects in automated lid-attach assembly; isolated needle geometry as root cause and validated improvements via optical metrology, reducing dispense error by 96%
- Designed automated engineered fluid dispensing (EFD) process replacing manual epoxy application for scalable production; validated process equivalence using cross-sectioning, optical measurement, and JEDEC MSL testing

Texas Inventionworks – Austin, TX

January 2024 – Present

Vice President • Prior Roles: Operations Executive Lead, Student Technician

- Led 60+ staff across operations, training, and technical services supporting 3,000+ students; scaled access to manufacturing and rapid prototyping resources including machining, additive manufacturing, and laser cutting
- Developed and taught mechanical and electrical hardware workshops guiding students through full product design cycles including CAD modeling, prototype fabrication, design validation testing, and design iteration

Nanoscale Design and Manufacturing Laboratory – Austin, TX

June 2024 – January 2025

Undergraduate Research Assistant – Micro-scale Selective Laser Sintering

- Designed SLA resin reservoir for leak-free microparticle suspension storage, selecting resin over FDM for non-porous sealing properties and simplified fabrication; reduced refill time and enabled faster setup time
- Developed Python-LabVIEW integrations to automate sinter pattern generation and synchronize DMD output; improved process consistency and reduced setup time through standardization and automation

DESIGN PROJECTS

National Medical Device Make-A-Thon – San Antonio, TX

February 2024

Second Place – Women's Health Challenge

- Identified redesign opportunity for gynecological tool through research on patient discomfort; analyzed prior art and existing improvement efforts to inform design; developed SolidWorks model and FDM-printed functional prototype

SKILLS

Mechanical Design & Fabrication: Concept Ideation, Human-Centered Design, DFM/DFA, Rapid Prototyping, Machining, Laser Cutting, Soldering, Tolerance Analysis, GD&T, EFD tooling, Carbon Fiber Assembly

Process & Manufacturing: DoE, Lean Principles, 5S methodology, A3 problem solving, Root Cause Analysis, Failure Analysis

Software: SolidWorks, Python, LabVIEW, MATLAB, Java (Oracle SE 8 Certified), Jira, Microsoft Suite